

The Unreliable Recipe

Science · Stage 4 (Year 8) · Chemical Science (Mixtures & Separation)

TASK DESCRIPTION

Students are given a 'recipe' for a mixture (e.g. a trail mix or a cleaning solution) that a fictional company wants scaled up for mass production — but the recipe has an ingredient that doesn't actually mix properly at scale. Students must identify the flaw, design a separation method to fix it, and pitch their fix to 'the company' (a short justification, not a lab report).

WILLIAMS MODEL PROCESSES TAGGED



WHY IT SUITS UNDERACHIEVING GIFTED STUDENTS

Open-ended entry point with no single correct method, so there is no ceiling on how far a student can take it. A real audience and product removes the 'pointless worksheet' feeling that often disengages these students. The task forces justification of a choice rather than simply producing an answer, which builds the work-strategy skills these students often haven't needed to develop before.

Draft example card — used to test the task-bank card format for the Gifted & Underachieving resource site (Science, Stage 4 & 5 template).